INCH POUND
MIL-PRF-1/343C
16 July 1998
SUPERSEDING
MIL-E-1/343B
4 February 1972

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING
TYPES 3Q4 AND 3V4

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tubes described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Power amplifier pentode.

Outline - - - 5-2 (EIA)

Base - - - E7-1

Envelope - - - T5-1/2

Cathode - - - Coated filament

Base connections:

Pin no. Element	1	2	3	4	5	6	7
3Q4	-f (+prl)	a	g1	g2	fct g3 (-prl)	a	+f
3V4	-f (+prl)	a	g2	nc	fct g3 (-prl)	g1	+f

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ik
Unit:	V dc	V dc	V dc	V dc	mA dc
Maximum:	1.6 (prl) or 3.2 (ser)	100		100	13
Minimum:	1.2 (prl) or 2.4 (ser)	- - -	- - -	- - -	- - -
Test conditions:	1.4 (prl)	90	-4.5	90	- - -

GENERAL:

Qualification - Not required.

DISTRIBUTION STATEMENT A. Approved for pubic release; distribution is unlimited.

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TABLE I. Testing and inspection.

Inspection	Method	Notes	Conditions	Acceptance level	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
Conformance inspection, part 1									
Total grid current	1266	2		1.0	II	Ic	0	-1.0	µA dc
Power output (2)	1341	2	Esig = 3.2 Vac; Ef = 1.1 V dc (prl); Rp = 0.01 Meg	1.0	II	Po	140	- - -	mW
Electrode current (anode)	1256	- - -		1.0	II	Ib	6.5	12.5	mA dc
Audio frequency noise	1246	1	Ecal (0 vu) = 2.63 Vac; Ecal (50 mW) = 400 mV dc; Rg1 = 0.1 meg; Rg2 = 1,000 Ω; Ecc2 = 55 V dc; Rp = 25,000 Ω; Ef = 2.8 V dc	1.0	II	EB	- - -	0	vu
Short and discontinuity detection	1201	- - -	40 G	0.4	II	- - -	- - -	- - -	- - -
Conformance inspection, part 2									
Filament current	1301	- - -		- - -	- - -	If	88	112	mA dc
Electrode current (screen grid)	1256	- - -		- - -	- - -	Ic2	1.3	3.1	mA dc
Transconductance	1306	- - -		- - -	- - -	Sm	1,800	2,500	µmhos
Power output (1)	1341	- - -	Esig = 3.2 Vac; Rp = 0.01 Meg	- - -	- - -	Po	210	- - -	mW
Direct-interelectrode capacitance	1331	- - -							
3Q4		- - -		- - -	- - -	Cg1-p	- - -	0.41	pF
3V4		- - -		- - -	- - -	Cg1-p	- - -	0.40	pF
3Q4		- - -	Without shield	- - -	- - -	Cin	3.8	7.2	pF
3V4		- - -		- - -	- - -	Cin	3.8	7.3	pF
3Q4		- - -		- - -	- - -	Cout	2.4	5.6	pF
3V4		- - -		- - -	- - -	Cout	2.2	5.4	pF
Low-frequency vibration	1031	- - -	Rp = 2,000 Ω	- - -	- - -	Ep	- - -	400	mV ac
Insulation of electrodes	1211	- - -		- - -	- - -	- - -	- - -	- - -	- - -
Base strain	1121	- - -		- - -	- - -	- - -	- - -	- - -	- - -
Glass strain	2126	- - -		- - -	- - -	- - -	- - -	- - -	- - -
Permanence of marking	1105	- - -		- - -	- - -	- - -	- - -	- - -	- - -

See footnotes at end of table I.

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TABLE 1. Testing and inspection. - Continued.

Inspection	Method	Notes	Conditions	Acceptance level	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
Conformance inspection, part 3									
Intermittent life	1501	- - -	Group A	- - -	- - -	- - -	- - -	- - -	- - -
Intermittent life-test end point (500 hours)	- - -								
Power output (1)	1341	- - -		- - -	- - -	Po	135	- - -	mW

NOTES:

1. See “Reduced pressure (altitude) rating”, and altitude, maximum peak voltage in the basic document.
2. This test to be performed at the conclusion of the holding period.
3. Ebb = Ecc2 = 135 V dc; Ec1 = 0; Rg1 = 3.3 Meg; Rp = 0.27 Meg; Rg2 = 2 Meg (by-passed with a 0.5 µF capacitor to -f). Remove tube and adjust gain of amplifier for 50 mW output with Esig = 3.0 V ac.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Preparing activity:
DLA-CC

(Project 5960-3459)

Review activities:
Army - AV
Navy - AS, CG, MC, OS
Air Force - 11, 19, 99

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